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1. Problem Statement: Write a C program that prompts the user to enter an integer and then determines whether the entered number is even or odd.

```
#include <stdio.h>

int main() {
    int num;
    printf("Enter an integer: ");
    scanf("%d", &num);
    if (num % 2 == 0) {
        printf("%d is an even integer.\n", num);
    } else {
        printf("%d is an odd integer.\n", num);
    }
    return 0;
}
```

Use code with caution.

Output:

Enter an integer: 7

7 is an odd integer.

2. Problem Statement: Write a C program to calculate the sum of the first 'n' natural numbers, where 'n' is an integer input provided by the user.

```
#include <stdio.h>

int main() {
    int n, i, sum = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    for (i = 1; i <= n; ++i) {
        sum += i;
    }
    printf("Sum of first %d natural numbers is: %d\n", n, sum);
    return 0;
}
```

Use code with caution.

Output:

Enter a positive integer: 5

Sum of first 5 natural numbers is: 15

3. Problem Statement: Write a C program that takes three numbers as input from the user and displays the largest of the three.

```
#include <stdio.h>

int main() {
    float num1, num2, num3;
    printf("Enter three numbers: ");
    scanf("%f %f %f", &num1, &num2, &num3);
    if (num1 >= num2 && num1 >= num3) {
        printf("Largest Number = %.2f\n", num1);
    } else if (num2 >= num1 && num2 >= num3) {
        printf("Largest Number = %.2f\n", num2);
    } else {
        printf("Largest Number = %.2f\n", num3);
    }
    return 0;
}
```

Use code with caution.

Output:

Enter three numbers: 10.5

8.2

15.1

Largest Number = 15.10

4. Problem Statement: Write a C program to calculate the factorial of a non-negative integer entered by the user.

```
#include <stdio.h>

int main() {
    int n, i;
    unsigned long long factorial = 1; // use unsigned long long for larger results
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    if (n < 0) {
        printf("Error: Factorial of a negative number does not exist.\n");
    } else {
        for (i = 1; i <= n; ++i) {
            factorial *= i;
        }
        printf("Factorial of %d is %llu.\n", n, factorial);
    }
    return 0;
}
```

Use code with caution.

Output:

Enter a positive integer: 5

Factorial of 5 is 120.

5. Problem Statement: Write a C program that acts as a simple calculator. It should take two operands and an operator (+, -, *, /) from the user and perform the corresponding calculation using a switch statement.

```
#include <stdio.h>

int main() {

    char op;

    double num1, num2, result;

    printf("Enter operator (+, -, *, /): ");

    scanf(" %c", &op);

    printf("Enter two operands: ");

    scanf("%lf %lf", &num1, &num2);

    switch (op) {

        case '+':

            result = num1 + num2;

            printf("%.1lf + %.1lf = %.1lf\n", num1, num2, result);

            break;

        case '-':

            result = num1 - num2;

            printf("%.1lf - %.1lf = %.1lf\n", num1, num2, result);

            break;

        case '*':

            result = num1 * num2;

            printf("%.1lf * %.1lf = %.1lf\n", num1, num2, result);

            break;

        case '/':

            if (num2 != 0) {

                result = num1 / num2;

                printf("%.1lf / %.1lf = %.1lf\n", num1, num2, result);

            } else {

                printf("Error: Division by zero is not allowed.\n");

            }

    }

}
```

```
        break;
    default:
        printf("Error: Invalid operator.\n");
    }
    return 0;
}
```

Use code with caution.

Output:

Enter operator (+, -, *, /): *

Enter two operands: 2.5

4.0

2.5 * 4.0 = 10.0